	1	2	3	4	5	6	7	8	9	12	14	18	27	30	31	37	40	71	76	112	419	1496
1	2717	312	102	36	80	48	0	64	108	0	0	0	0	0	0	0	0	0	0	0	0	0
2	1174	164	156	144	0	48	0	0	0	0	0	0	0	0	248	0	0	0	0	0	0	0
3	813	378	225	48	30	36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	328	64	24	32	0	48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	410	200	120	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	606	228	234	96	0	216	0	96	0	72	0	0	0	180	0	0	0	0	0	0	0	0
7	70	252	84	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	512	80	240	0	0	0	0	64	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	486	216	54	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	252	72	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	14	56	0	0	350	0	196	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	72	0	108	144	0	0	0	0	0	0	0	0	0	0	0	0	1440	0	0	0	0	0
27	216	0	324	0	0	0	0	0	0	0	0	0	0	0	0	0	2160	0	0	0	0	0
30	510	300	0	0	0	0	0	480	0	0	0	0	0	0	0	0	0	0	0	0	0	0
31	62	372	372	0	0	0	0	992	0	0	0	0	0	0	0	0	0	0	0	0	0	0
37	370	296	0	0	0	0	0	0	0	0	0	1332	0	0	0	0	0	0	0	0	0	0
40	240	320	0	0	0	0	0	640	0	0	0	0	0	0	0	0	0	0	0	0	0	0
71	1704	0	426	568	0	0	0	0	0	0	0	0	0	0	0	5254	0	0	10792	15904	0	0
76	760	608	912	608	0	0	0	0	0	0	0	0	0	0	0	0	6080	0	0	0	0	0
112	2016	1792	672	0	0	1344	1568	0	2016	0	0	4032	6048	0	0	0	0	0	0	0	0	0
419	4609	1676	1257	0	0	2514	0	0	0	0	5866	0	0	0	0	31006	0	0	0	0	0	626824
1496	59840	0	4488	5984	0	8976	0	0	0	0	0	0	0	0	0	0	0	212432	0	0	0	0

Table 1: Depicts the number of strict edges from each reversible edge to another.

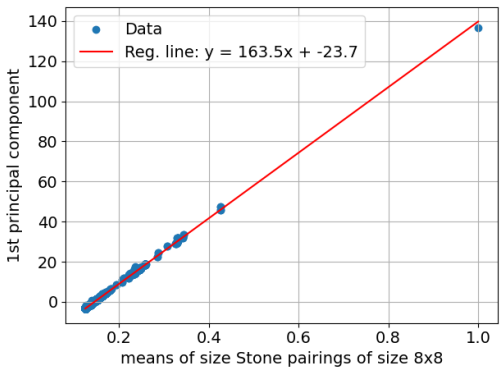

Figure 8: Linear regression graph of the expectation (or mean) of the Stone pairing for each equational theory with relation to the first principal component X.

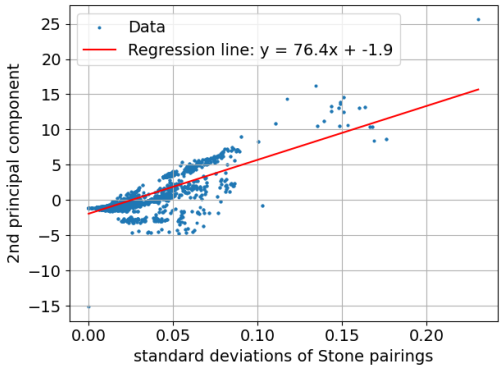

Figure 9: Regression graph of the standard value of the Stone pairings for each equation vrs. the second principal component Y.

B Cliques of provably equivalent theories

The repartition of reversible cliques follows:

cardinal of the clique	number of cliques of this cardinal	number of vertices
1496	1	1496
419	1	419
112	2	224
76	2	152
71	2	142
40	2	80
37	2	74
31	2	62
30	1	30
27	2	54
18	2	36
14	1	14
12	1	12
9	4	36
8	7	56
7	4	28
6	11	66
5	11	55
4	14	56
3	61	183
2	137	274
1	1145	1145

size of clique	selected equation in the clique	
1496	Eqn[1]	$x = y$
419	Eqn[45]	$x \diamond y = z \diamond w$
112	Eqn[18]	$x = y \diamond (z \diamond x)$
112	Eqn[26]	$x = (x \diamond y) \diamond z$
76	Eqn[93]	$x = y \diamond (z \diamond (w \diamond x))$
76	Eqn[268]	$x = ((x \diamond y) \diamond z) \diamond w$
71	Eqn[3]	$x = x \diamond y$
71	Eqn[4]	$x = y \diamond x$
40	Eqn[41]	$x \diamond y = x \diamond z$
40	Eqn[44]	$x \diamond y = z \diamond y$
37	Eqn[354]	$x \diamond y = z \diamond (w \diamond y)$
37	Eqn[382]	$x \diamond y = (x \diamond z) \diamond w$
31	Eqn[4378]	$x \diamond (y \diamond z) = w \diamond (u \diamond v)$
31	Eqn[4693]	$(x \diamond y) \diamond z = (w \diamond u) \diamond v$
30	Eqn[710]	$x = y \diamond (y \diamond ((x \diamond z) \diamond z))$
	Eqn[2943]	$x = ((y \diamond (y \diamond x)) \diamond z) \diamond z$
27	Eqn[607]	$x = y \diamond (z \diamond (w \diamond (u \diamond x)))$
27	Eqn[3100]	$x = (((x \diamond y) \diamond z) \diamond w) \diamond u$
18	Eqn[3450]	$x \diamond y = z \diamond (w \diamond (u \diamond y))$
18	Eqn[4152]	$x \diamond y = ((x \diamond z) \diamond w) \diamond u$
14	Eqn[4581]	$x \diamond (y \diamond z) = (w \diamond u) \diamond v$
12	Eqn[1354]	$x = y \diamond (((z \diamond x) \diamond y) \diamond z)$
	Eqn[2369]	$x = (y \diamond (z \diamond (x \diamond y))) \diamond z$
9	Eqn[8]	$x = x \diamond (x \diamond y)$
9	Eqn[27]	$x = (y \diamond x) \diamond x$
9	Eqn[1815]	$x = (y \diamond z) \diamond ((w \diamond z) \diamond x))$
9	Eqn[1873]	$x = (x \diamond (y \diamond z)) \diamond (y \diamond w)$
8	Eqn[13]	$x = y \diamond (x \diamond y)$
	Eqn[28]	$x = (y \diamond x) \diamond y$
8	Eqn[745]	$x = y \diamond (z \diamond (x \diamond y) \diamond z)$
	Eqn[2978]	$x = ((y \diamond (z \diamond x)) \diamond y) \diamond z$
8	Eqn[3369]	$x \diamond y = y \diamond (z \diamond (z \diamond x))$
	Eqn[4181]	$x \diamond y = ((y \diamond z) \diamond z) \diamond x$
8	Eqn[4360]	$x \diamond (y \diamond z) = x \diamond (w \diamond u)$
8	Eqn[4692]	$(x \diamond y) \diamond z = (w \diamond u) \diamond z$
8	Eqn[4377]	$x \diamond (y \diamond z) = w \diamond (u \diamond z)$
8	Eqn[4675]	$(x \diamond y) \diamond z = (x \diamond w) \diamond u$
7	Eqn[160]	$x = (x \diamond y) \diamond (y \diamond z)$
7	Eqn[193]	$x = (y \diamond z) \diamond (z \diamond x)$
7	Eqn[4467]	$x \diamond (y \diamond x) = (z \diamond w) \diamond u$
7	Eqn[4579]	$x \diamond (y \diamond z) = (w \diamond u) \diamond w$

Figure 10: List of all equivalence classes of equational theories modulo $\sim$, of size between 7 and 1496, together with one or two distinctive elements of each clique.

C Herbrand theorem and parallelism of implication flows

Suppose given a pair of magma equations

$$\varphi(\mathbf{u}) = \varphi(u_1, \ldots, u_k)$$

where φ has free variables $\mathbf{u} = (u_1, \ldots, u_k)$ and

$$\psi(\mathbf{x}) = \psi(x_1, \ldots, x_\ell).$$

where ψ has free variables $\mathbf{x} = (x_1, \ldots, x_\ell)$.

A Herbrand proof

$$(\theta_1, \ldots, \theta_n) \quad : \quad \forall \mathbf{u}.\varphi(\mathbf{u}) \ \vdash \ \forall \mathbf{x}.\psi(\mathbf{x})$$

is defined as a finite sequence $(\theta_1, \ldots, \theta_n)$ where each θ_i is a substitution of the free variables

$$\mathbf{u} = (u_1, \ldots, u_k)$$

of the equation φ with magma expressions parametrized by the free variables

$$\mathbf{x} = (x_1, \ldots, x_\ell)$$

of the equation ψ, such that one can establish

$$\forall \mathbf{x}. \ \Big(\ \varphi[\theta_1] \wedge \cdots \wedge \varphi[\theta_n] \ \Big) \ \Rightarrow \ \phi(\mathbf{x}).$$

Consider for instance Eqn[13]

$$\forall \mathbf{u}. \, \varphi(\mathbf{u}) \quad \equiv \quad \forall u.v. \ u = v \diamond (u \diamond v)$$

and Eqn[1557]

$$\forall \mathbf{x}. \, \psi(\mathbf{x}) \ \equiv \ \forall x.y.z. \ x = (y \diamond z) \diamond (x \diamond (y \diamond z))$$

with free variables $\mathbf{u} = (u, v)$ and $\mathbf{x} = (x, y, z)$, respectively. In one direction, the fact that Eqn[13] implies Eqn[1557] is obvious by substituting the two variables u and v with the expressions

$$\theta : u \mapsto x \quad \text{and} \quad \theta : v \mapsto x \diamond y.$$

From this, one concludes that the substitution θ establishes the implication:

$$\theta \quad : \quad \mathsf{Eqn}[13] \ \vdash \ \mathsf{Eqn}[1557]$$

Conversely, somewhat surprisingly, the Equational Theories project indicates that Eqn[1557] implies Eqn[13]. This nontrivial fact can be established by applying the pair of substitutions

$$\theta_1 \ : \ x \mapsto u, \quad y \mapsto u \diamond v, \quad z \mapsto v \diamond (u \diamond u)$$

$$\theta_2 \ : \ x \mapsto v, \quad y \mapsto u, \quad z \mapsto u.$$

to the equation

$$\psi(x, y, z) \quad \equiv \quad x \ = \ (y \diamond z) \diamond (x \diamond (y \diamond z))$$

in order to obtain the equation $\psi[\theta_1]$ after substitution by θ_1:

$u = ((u \diamond u) \diamond (v \diamond (u \diamond u))) \diamond (u \diamond ((u \diamond u) \diamond (v \diamond (u \diamond u))))$

and the equation $\psi[\theta_2]$ after substitution by θ_2:

$$v = (u \diamond u) \diamond (v \diamond (u \diamond u))$$

An easy equational reasoning establishes that the pair of the two equations implies $\mathsf{Eqn}[13]$:

$$\varphi(u, v) \quad \equiv \quad u = v \diamond (u \diamond v) \tag{2}$$

From this, one concludes that

$$(\theta_1, \theta_2) \quad : \quad \mathsf{Eqn}[1557] \ \vdash \ \mathsf{Eqn}[13]$$

Note that the two substitutions θ_3 and θ_4 defined below would establish (2) in just the same way:

$$\theta_3 \ : \ x \mapsto u, \quad y \mapsto u \diamond v, \quad z \mapsto v \diamond (u \diamond u)$$

$$\theta_4 \ : \ x \mapsto v, \quad y \mapsto u, \quad z \mapsto v.$$

From this follows that

$$(\theta_3, \theta_4) \quad : \quad \mathsf{Eqn}[1557] \ \vdash \ \mathsf{Eqn}[13]$$

This defines a category with equations as objects and Herbrand proofs as morphisms. The fact that $(\theta_1, \theta_2) \neq (\theta_3, \theta_4)$ means that this is a category and not just a preorder.

There are special commutative triangles:

$$\begin{tikzcd}
 & \forall \mathbf{u}.\psi(\mathbf{u}) \arrow[d, "\alpha"] & \forall \mathbf{x}.\phi(\mathbf{x}) \arrow[d, "\alpha"'] \arrow[dr, "\theta\circ\alpha"] & \\
\forall \mathbf{x}.\phi(\mathbf{x}) \arrow[ur, "\theta"] \arrow[dr, "\alpha\circ\theta"'] & & & \forall \mathbf{u}.\psi(\mathbf{u}) \\
 & \forall \mathbf{v}.\psi(\mathbf{v}) & \forall \mathbf{y}.\phi(\mathbf{y}) \arrow[ur, "\theta"'] &
\end{tikzcd}$$

where the sides are linear substitutions. These deserve to be filled, in order to define a higher-dimensional manifold (defined as a simplicial set) of equations and reasonings, constructed as a variation of the nerve of a category. We leave that for future work.